Recurrent Neural Network
RNN

Credit to Arun Mallya

Outline

- Why Recurrent Neural Networks (RNNs)?
- The Vanilla RNN unit
- The RNN forward pass
- Backpropagation refresher
- The RNN backward pass
- Issues with the Vanilla RNN
- The Long Short-Term Memory (LSTM) unit

Motivation

- Not all problems can be converted into one with fixed-length inputs and outputs

- Problems such as Speech Recognition or Time-series Prediction require a system to store and use context information
  - Simple case: Output YES if the number of 1s is even, else NO
    1000010101 – YES, 100011 – NO, ...

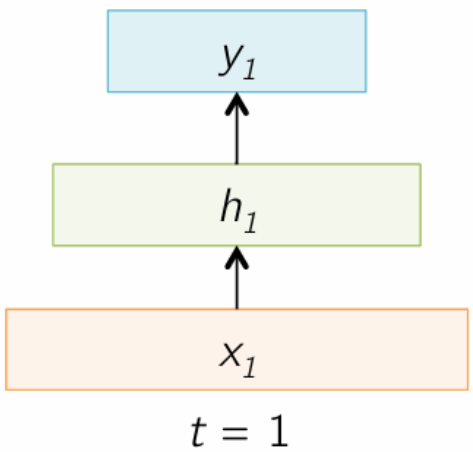
- Hard/Impossible to choose a fixed context window
  - There can always be a new sample longer than anything seen

Recurrent Neural Networks (RNNs)

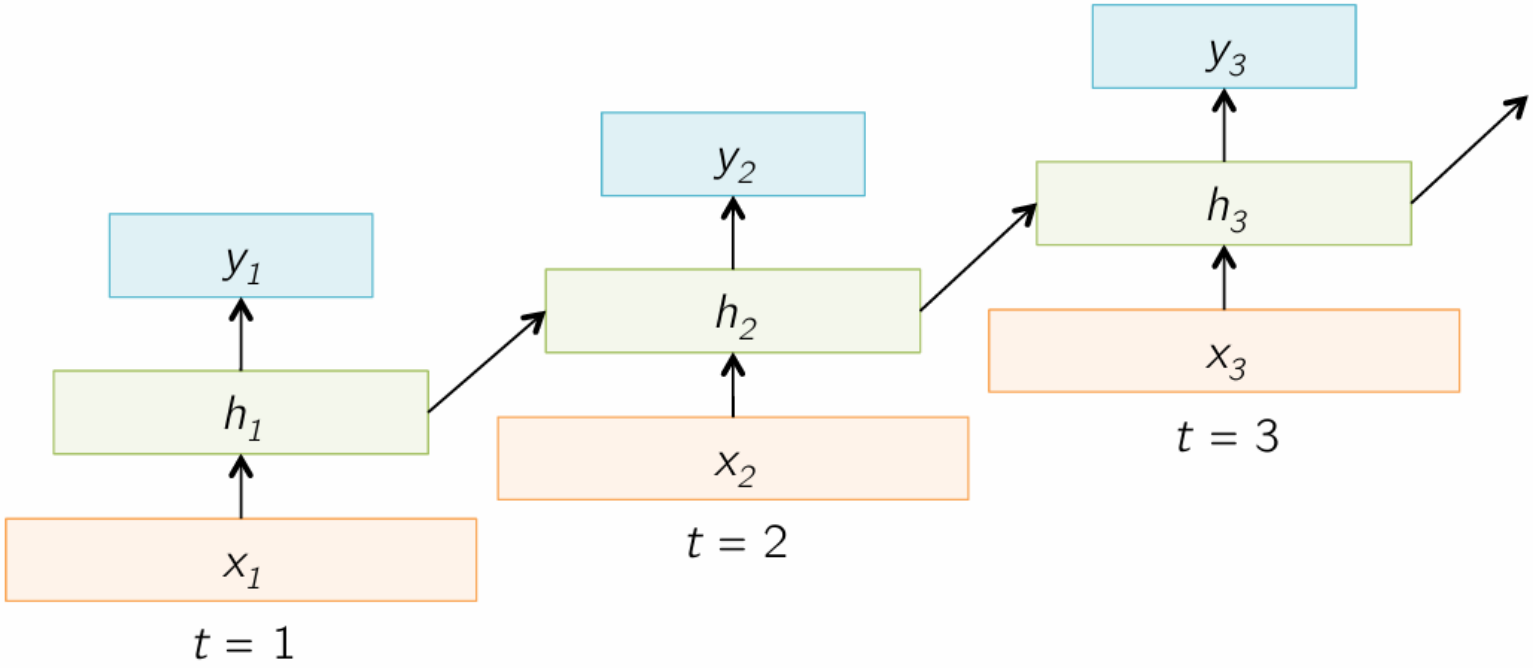
- **R**ecurrent **N**eural **N**etwork**s** take the previous output or hidden states as inputs.
  The composite input at time t has some historical information about the happenings at time $\mathrm{T} < t$

- RNNs are useful as their intermediate values (state) can store information about past inputs for a time that is not fixed a priori

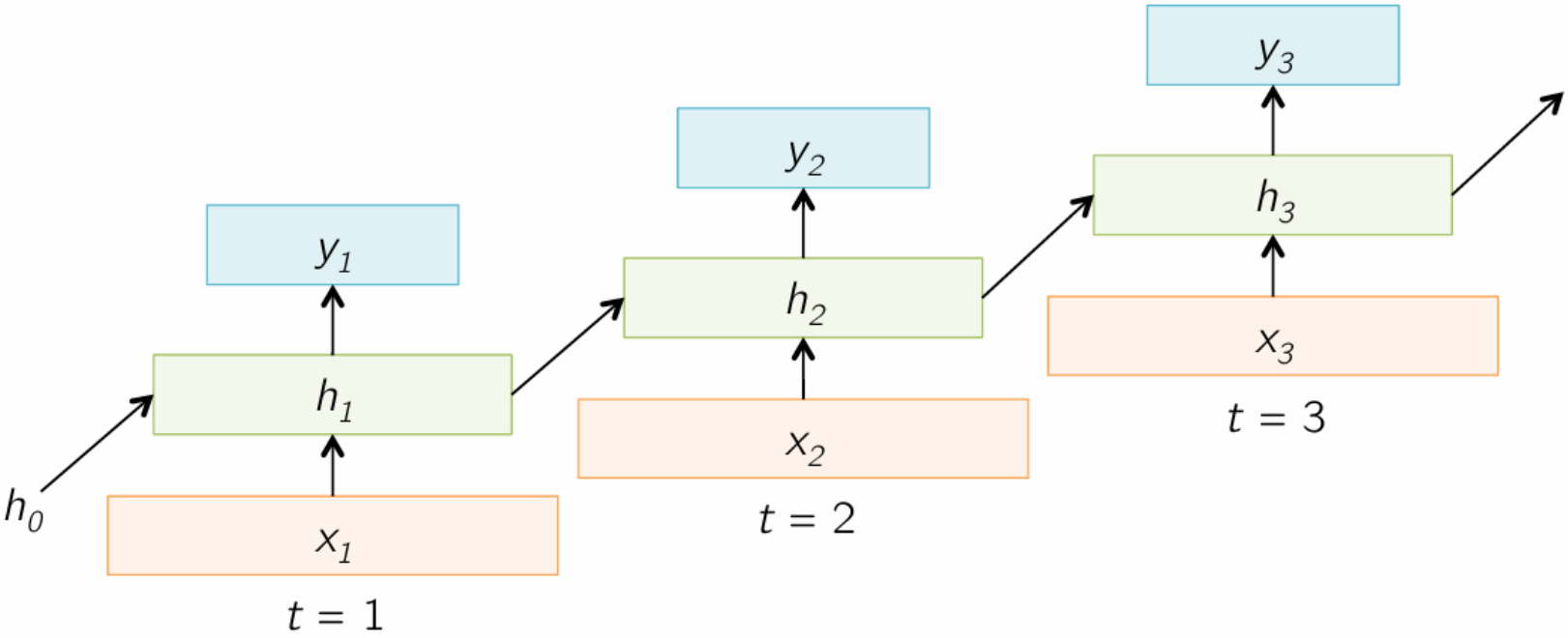
Sample Feed-forward Network



Sample RNN

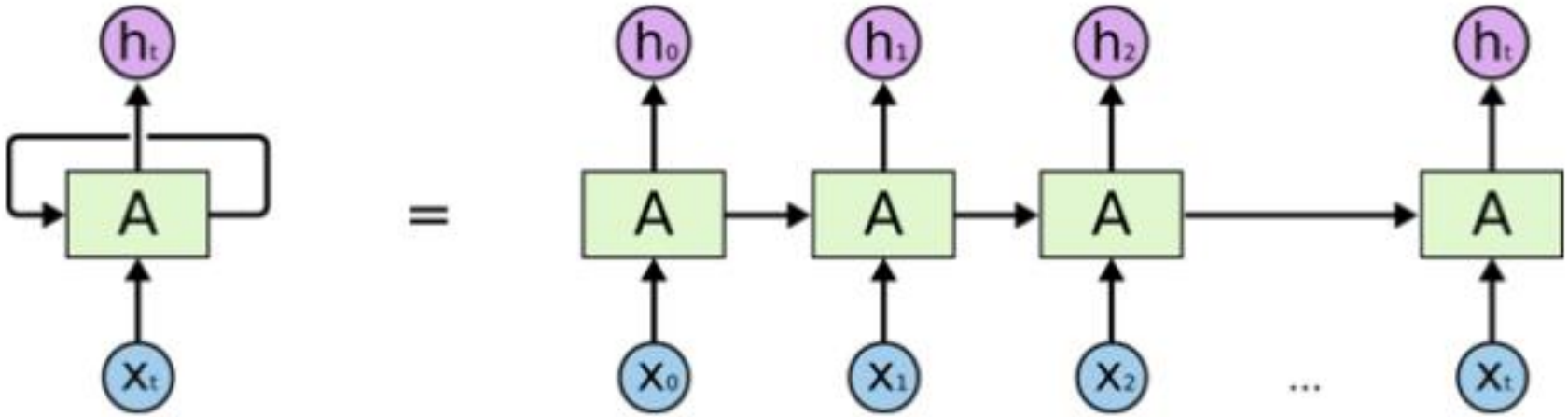


Sample RNN



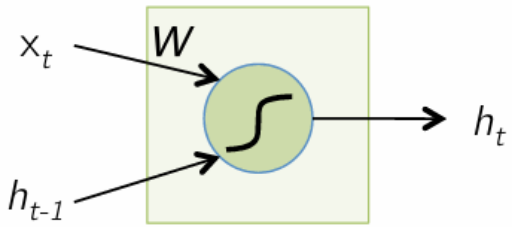
RNN unrolled

- Unroll and treat each time sample as an unit.



An unrolled RNN

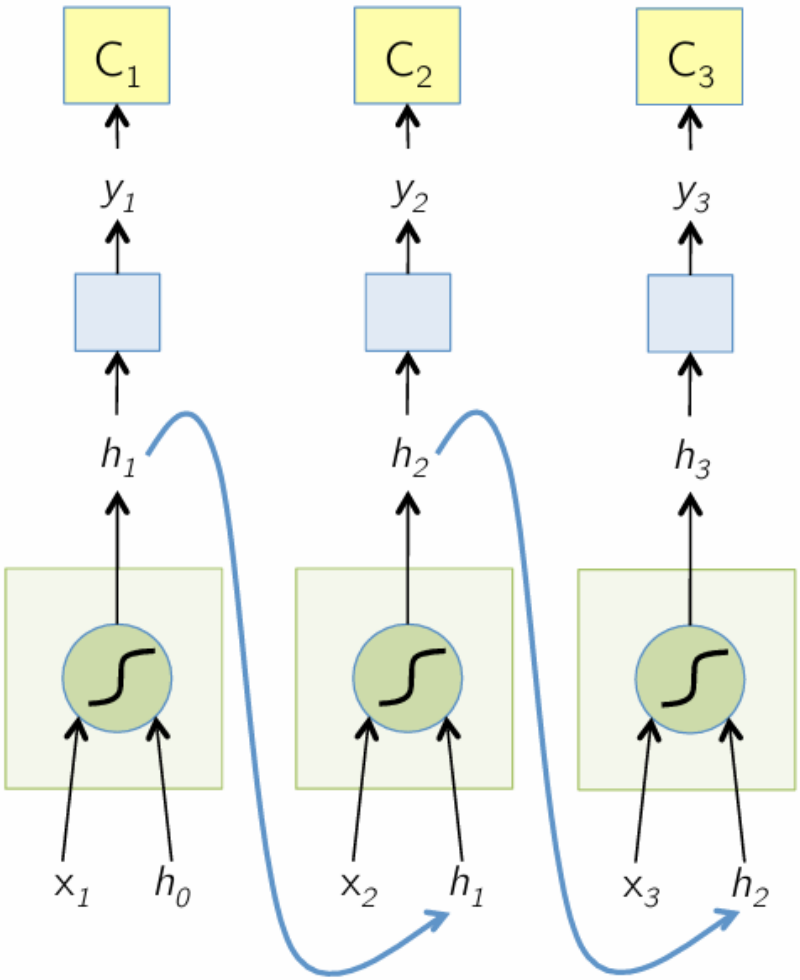
The Vanilla RNN Cell



$$h_t = \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

- x_t = input at time step **t.**
- h_{t-1} = previous hidden state (memory from the past).
- Both are concatenated into one vector.
- Multiplied by weight matrix W.
- Passed through **tanh activation function** to produce the new hidden state h_t.

The Vanilla RNN Forward



$$h_t = \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

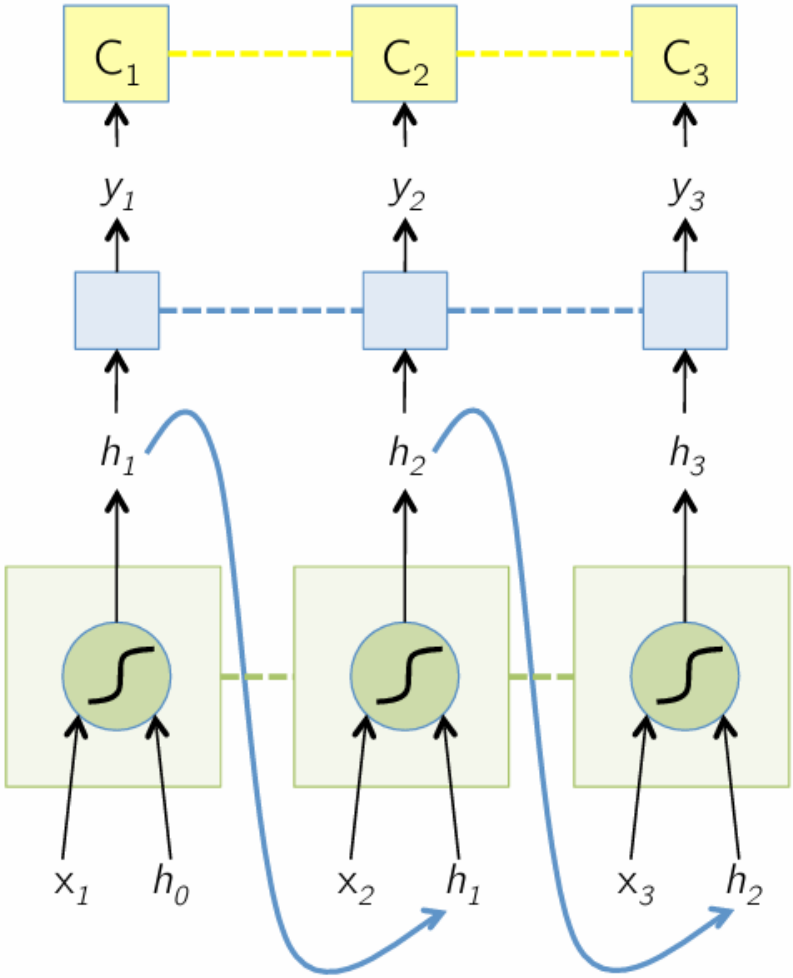
$$y_t = \mathrm{F}(h_t)$$

$$C_t = \mathrm{Loss}(y_t, \mathrm{GT}_t)$$

- The output at time t is computed from the hidden state.
- F is usually:
  - softmax (for classification)
  - linear layer (for regression)

- GT_t = ground truth (the correct label/output for time t)
- Compare predicted output y_t with the true value using a loss function.

The Vanilla RNN Forward



$$h_t = \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

$$y_t = \mathrm{F}(h_t)$$

$$C_t = \mathrm{Loss}(y_t, \mathrm{GT}_t)$$

------ indicates shared weights

Recurrent Neural Networks (RNNs)

- Note that the weights are shared over time
- Essentially, copies of the RNN cell are made over time (unrolling/unfolding), with different inputs at different time steps

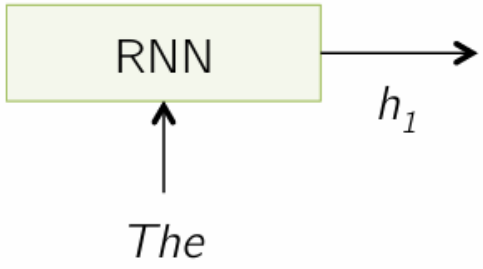
Sentiment Classification

- Classify a
  restaurant review from Yelp! OR
  movie review from IMDB OR
  ...
  as positive or negative

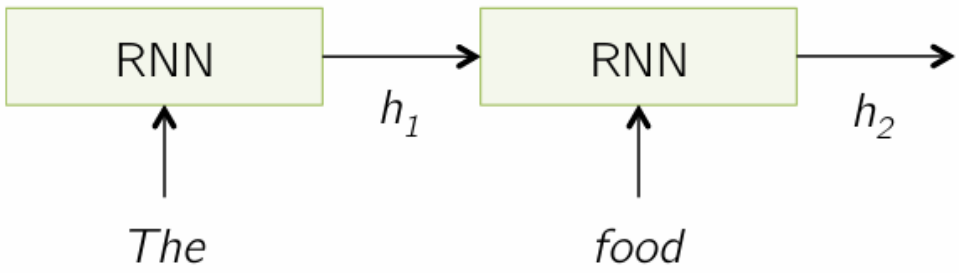
- **Inputs:** Multiple words, one or more sentences
- **Outputs:** Positive / Negative classification

- "The food was really good"

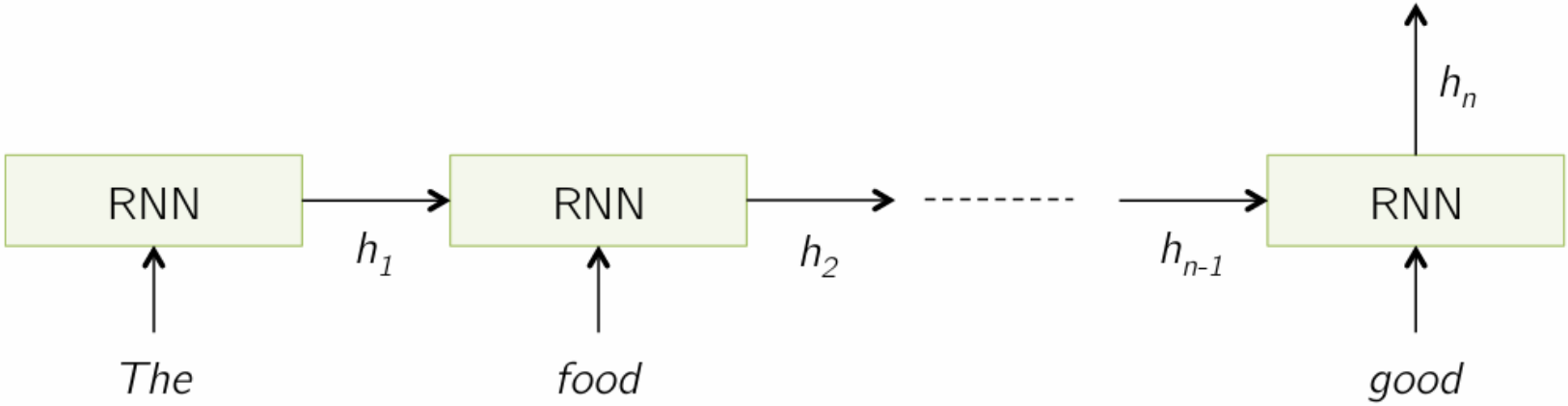
Sentiment Classification



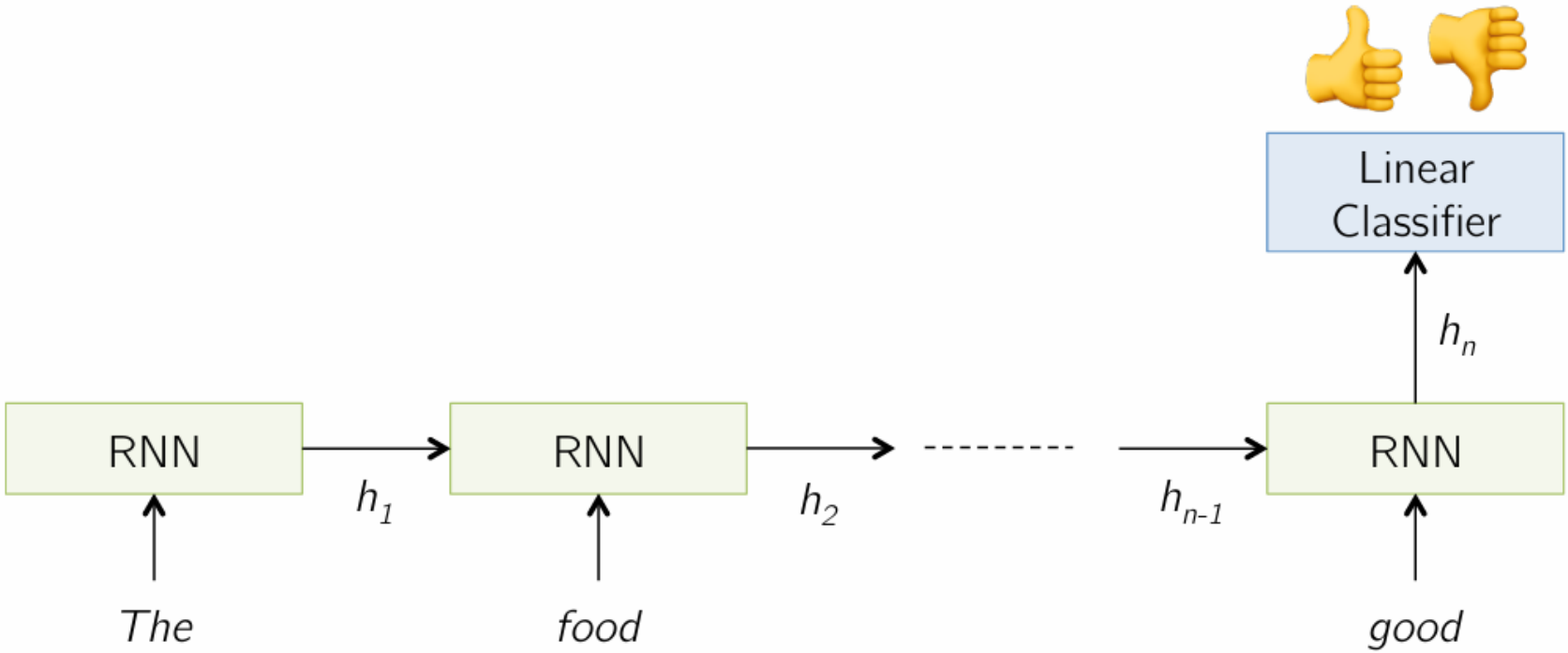
Sentiment Classification



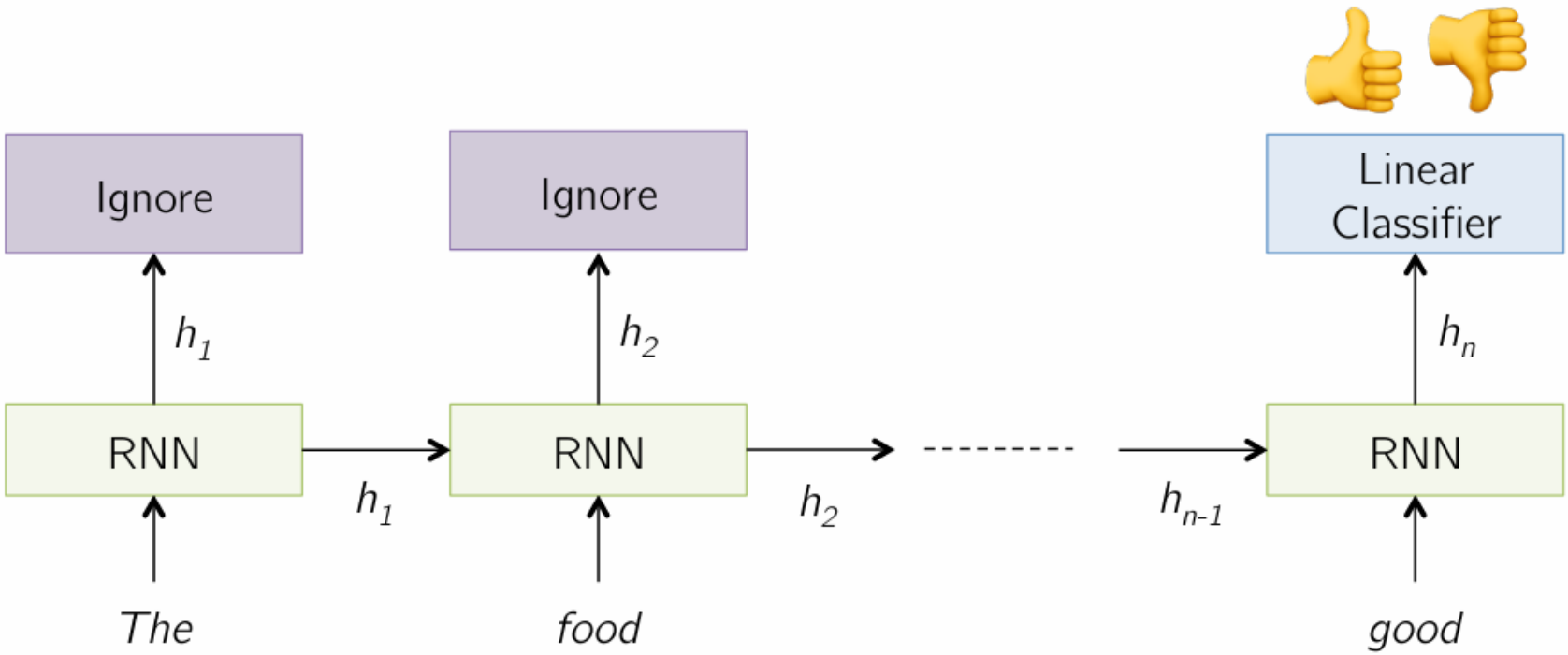
Sentiment Classification



Sentiment Classification



Sentiment Classification



Using ONLY the final hidden state (h_n)

You use the last hidden state as the sentence representation:

$$h = h_n$$

✓**Advantages:**

•Theoretically contains information from the entire sequence.

•Simple and fast.

•Works well for short sequences.

✕ **Disadvantages:**

•Earlier words may be **forgotten** (vanishing gradients).

•If the sentiment is at the beginning of the sentence, h_n may not capture it.

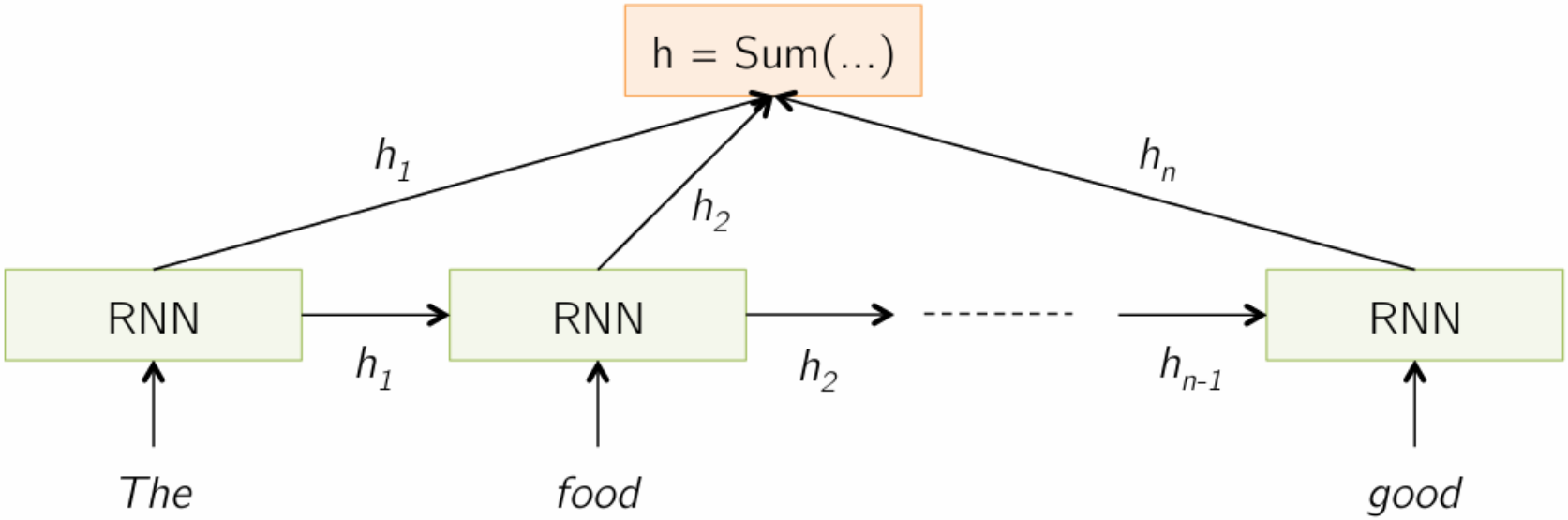
•Long sentences = worse performance.

Example where it fails:

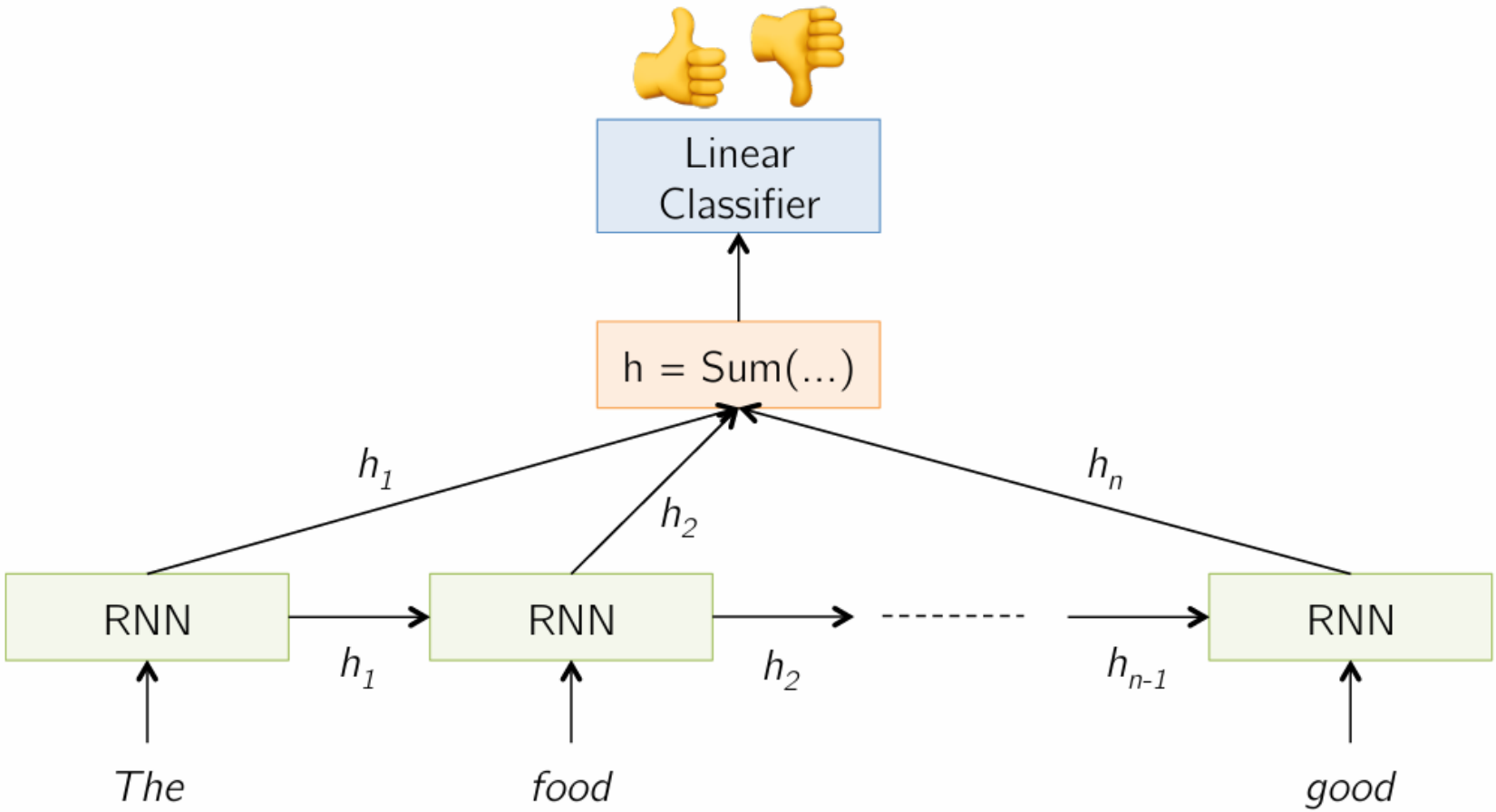
"I hated the appetizer, but the dessert was perfect!"

Final word "perfect" is positive → final hidden state may think the whole sentence is positive.

Sentiment Classification



Sentiment Classification



Using the SUM (or average) of ALL hidden states

✓**Advantages:**

- Uses **information from every word**.
- Not biased toward the final part of the sentence.
- Helps when sentiment is distributed across the whole text.
- Example:
- "The service was bad, but the food was good."
- Hidden states:
- h_2 → "bad"
- h_5 → "good"
- By summing all states, the model captures **both** signals → more balanced.

✕ **Disadvantages:**

- All words treated equally (important and unimportant words mixed).
- Noise from irrelevant words ("the", "is", "very") gets added.
- No understanding of word importance.

Image Captioning

- Given an image, produce a sentence describing its contents

- **Inputs:** Image feature (from a CNN)
- **Outputs:** Multiple words (let's consider one sentence)

: The dog is hiding

Image Captioning

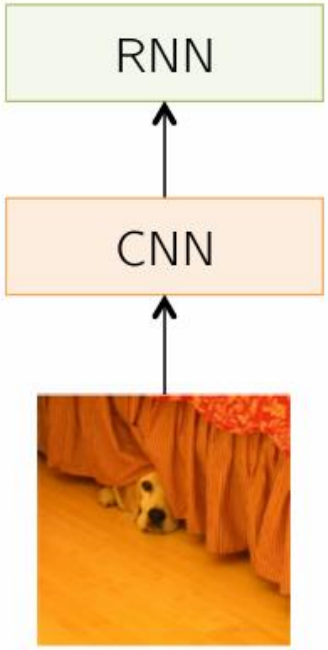


Image Captioning

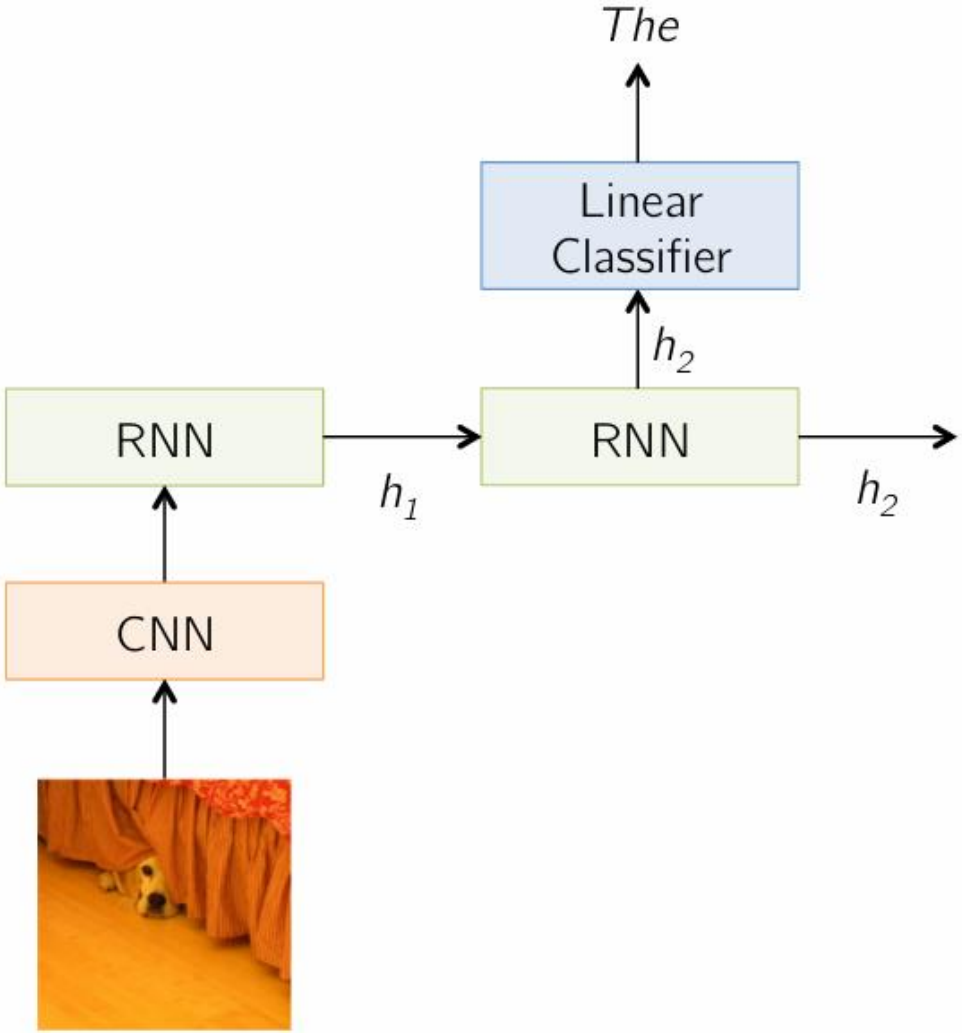


Image Captioning

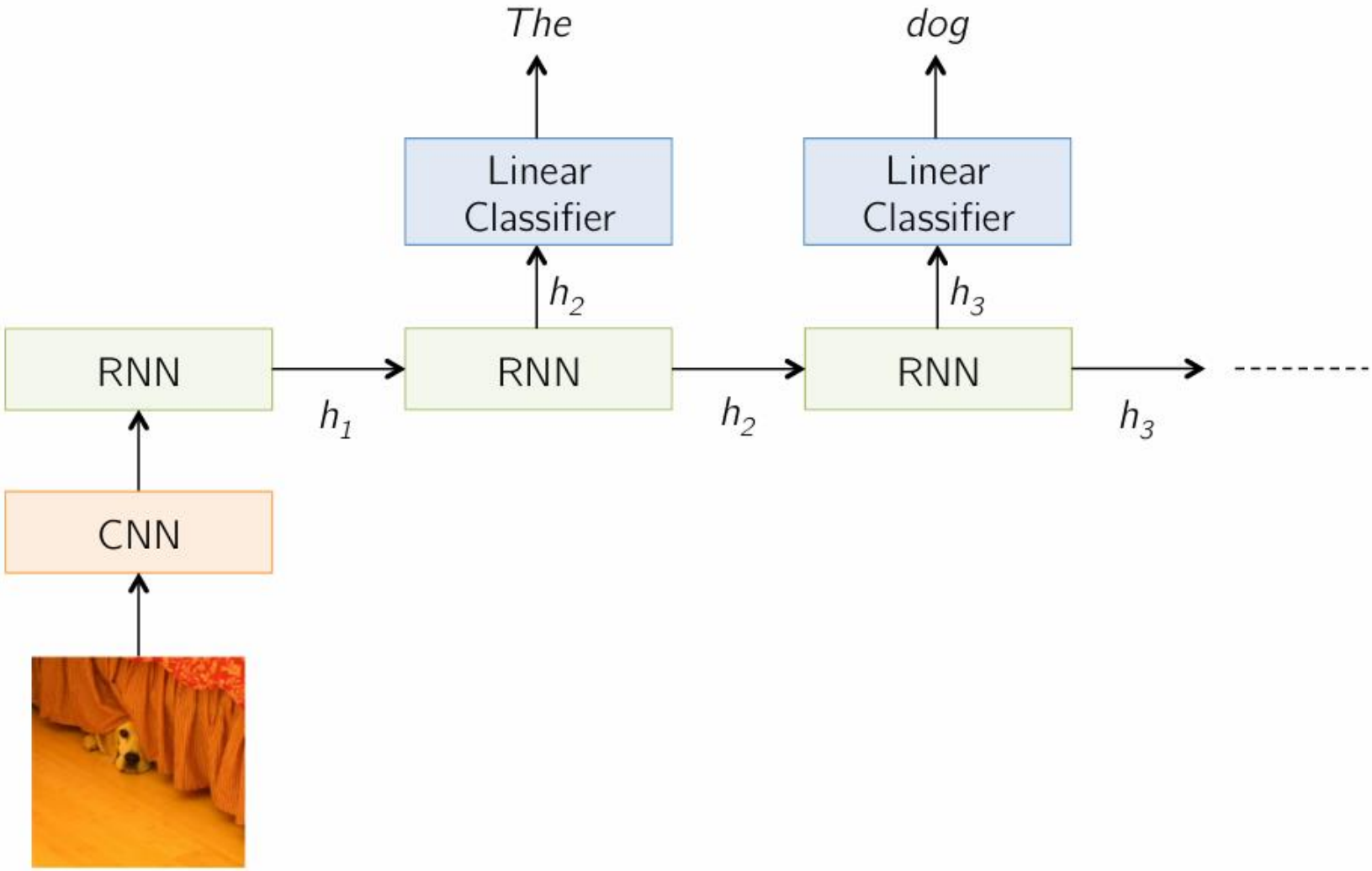


Image Captioning

✓ **CNN** extracts visual meaning
✓ **RNN** uses image meaning to start language generation
✓ **RNN** generates a sentence one word at a time
✓ Linear classifier converts each **RNN** output into a word
Final output might look like:
“The dog is hiding.”

RNN Outputs: Image Captions

A person riding a motorcycle on a dirt road.

Two dogs play in the grass.

A herd of elephants walking across a dry grass field.

A group of young people playing a game of frisbee.

Two hockey players are fighting over the puck.

A close up of a cat laying on a couch.

RNN Outputs: Language Modeling

VIOLA:
Why, Salisbury must find his flesh and thought
That which I am not aps, not a man and in fire,
To show the reining of the raven and the wars
To grace my hand reproach within, and not a fair are hand,
That Caesar and my goodly father's world;
When I was heaven of presence and our fleets,
We spare with hours, but cut thy council I am great,
Murdered and by thy master's ready there
My power to give thee but so much as hell:
Some service in the noble bondman here,
Would show him to her wine.

KING LEAR:
O, if you were a feeble sight, the courtesy of your law,
Your sight and several breath, will wear the gods
With his heads, and my hands are wonder'd at the deeds,
So drop upon your lordship's head, and your opinion
Shall be against your honour.

http://karpathy.github.io/2015/05/21/rnn-effectiveness/

Input – Output Scenarios

Single - Single

Feed-forward Network

Single - Multiple

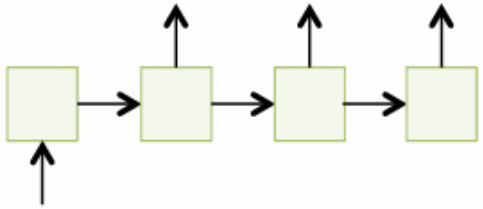
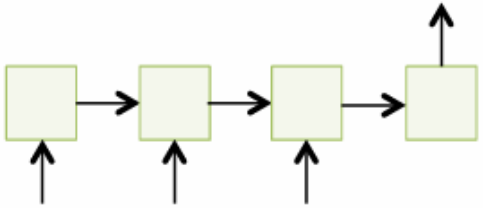


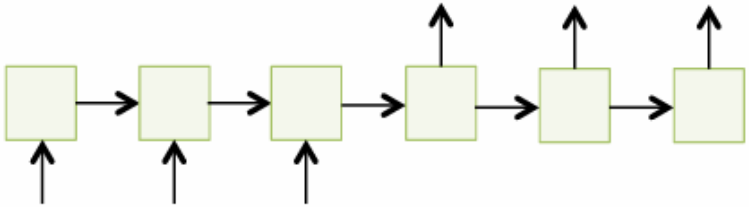
Image Captioning

Multiple - Single



Sentiment Classification

Multiple - Multiple



Translation

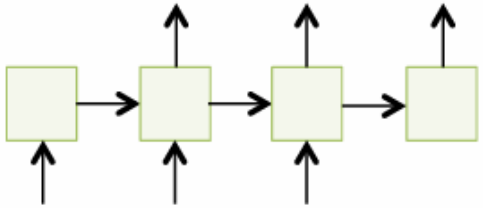


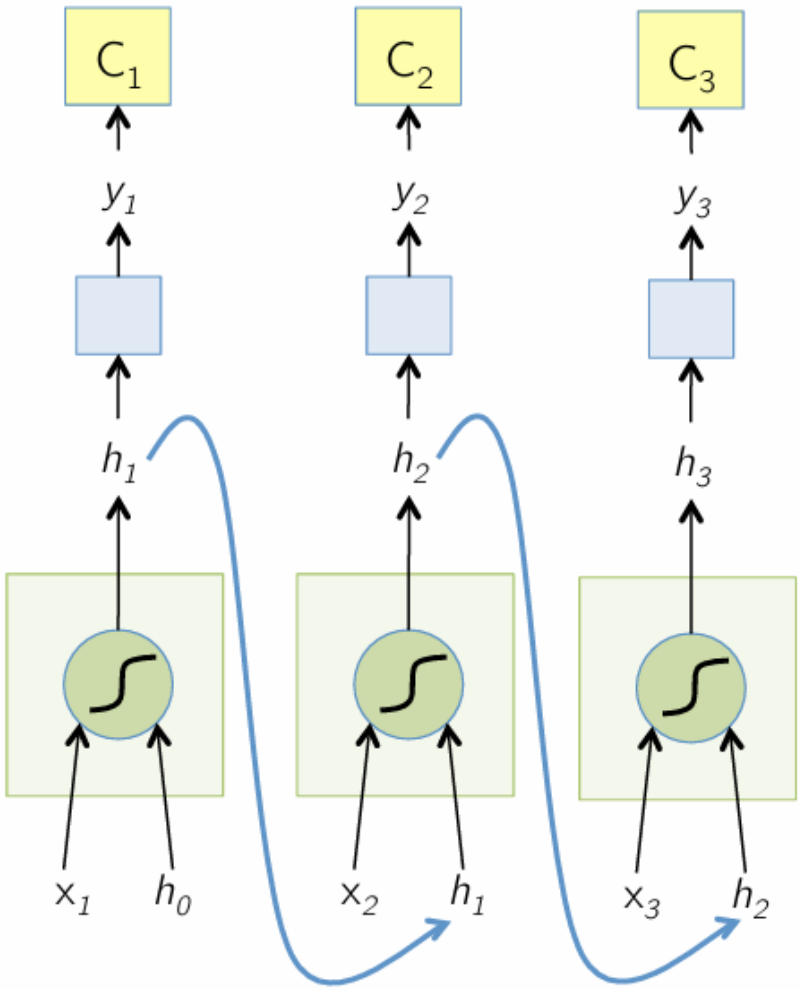
Image Captioning

Input – Output Scenarios

Note: We might deliberately choose to frame our problem as a particular input-output scenario for ease of training or better performance.

For example, at each time step, provide previous word as input for image captioning
(Single-Multiple to Multiple-Multiple).

The Vanilla RNN Forward



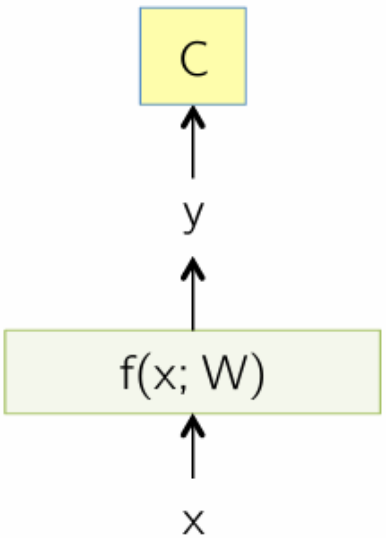
$$h_t = \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

$$y_t = \mathrm{F}(h_t)$$

$$C_t = \mathrm{Loss}(y_t, \mathrm{GT}_t)$$

"Unfold" network through time by making copies at each time-step

BackPropagation Refresher



$$y = f(x;W)$$

$$C = \text{Loss}(y, y_{GT})$$

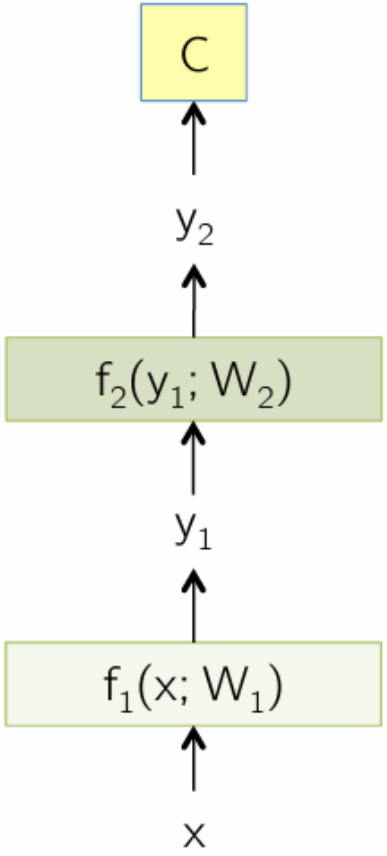
SGD Update

$$W \leftarrow W - \eta \frac{\partial C}{\partial W}$$

$$\frac{\partial C}{\partial W} = \left(\frac{\partial C}{\partial y}\right)\left(\frac{\partial y}{\partial W}\right)$$

- Stochastic Gradient Descent (SGD)

Multiple Layers



$$y_1 = f_1(x; W_1)$$

$$y_2 = f_2(y_1; W_2)$$

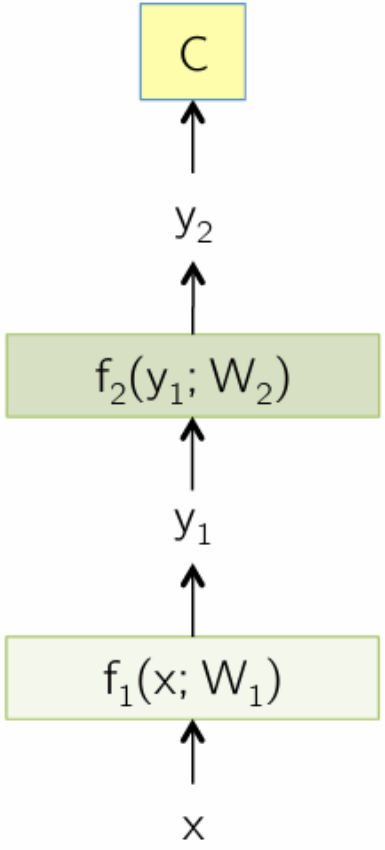
$$C = \text{Loss}(y_2, y_{GT})$$

SGD Update

$$W_2 \leftarrow W_2 - \eta \frac{\partial C}{\partial W_2}$$

$$W_1 \leftarrow W_1 - \eta \frac{\partial C}{\partial W_1}$$

Chain Rule for Gradient Computation



$$y_1 = f_1(x; W_1)$$

$$y_2 = f_2(y_1; W_2)$$

$$C = \text{Loss}(y_2, y_{GT})$$

$$\text{Find } \frac{\partial C}{\partial W_1}, \frac{\partial C}{\partial W_2}$$

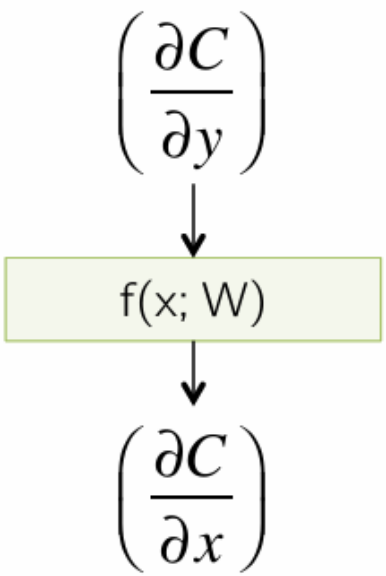
$$\frac{\partial C}{\partial W_2} = \left(\frac{\partial C}{\partial y_2}\right)\left(\frac{\partial y_2}{\partial W_2}\right)$$

$$\frac{\partial C}{\partial W_1} = \left(\frac{\partial C}{\partial y_1}\right)\left(\frac{\partial y_1}{\partial W_1}\right)$$

$$= \left(\frac{\partial C}{\partial y_2}\right)\left(\frac{\partial y_2}{\partial y_1}\right)\left(\frac{\partial y_1}{\partial W_1}\right)$$

Application of the Chain Rule

Chain Rule for Gradient Computation



Given: $\left(\frac{\partial C}{\partial y}\right)$

We are interested in computing: $\left(\frac{\partial C}{\partial W}\right), \left(\frac{\partial C}{\partial x}\right)$

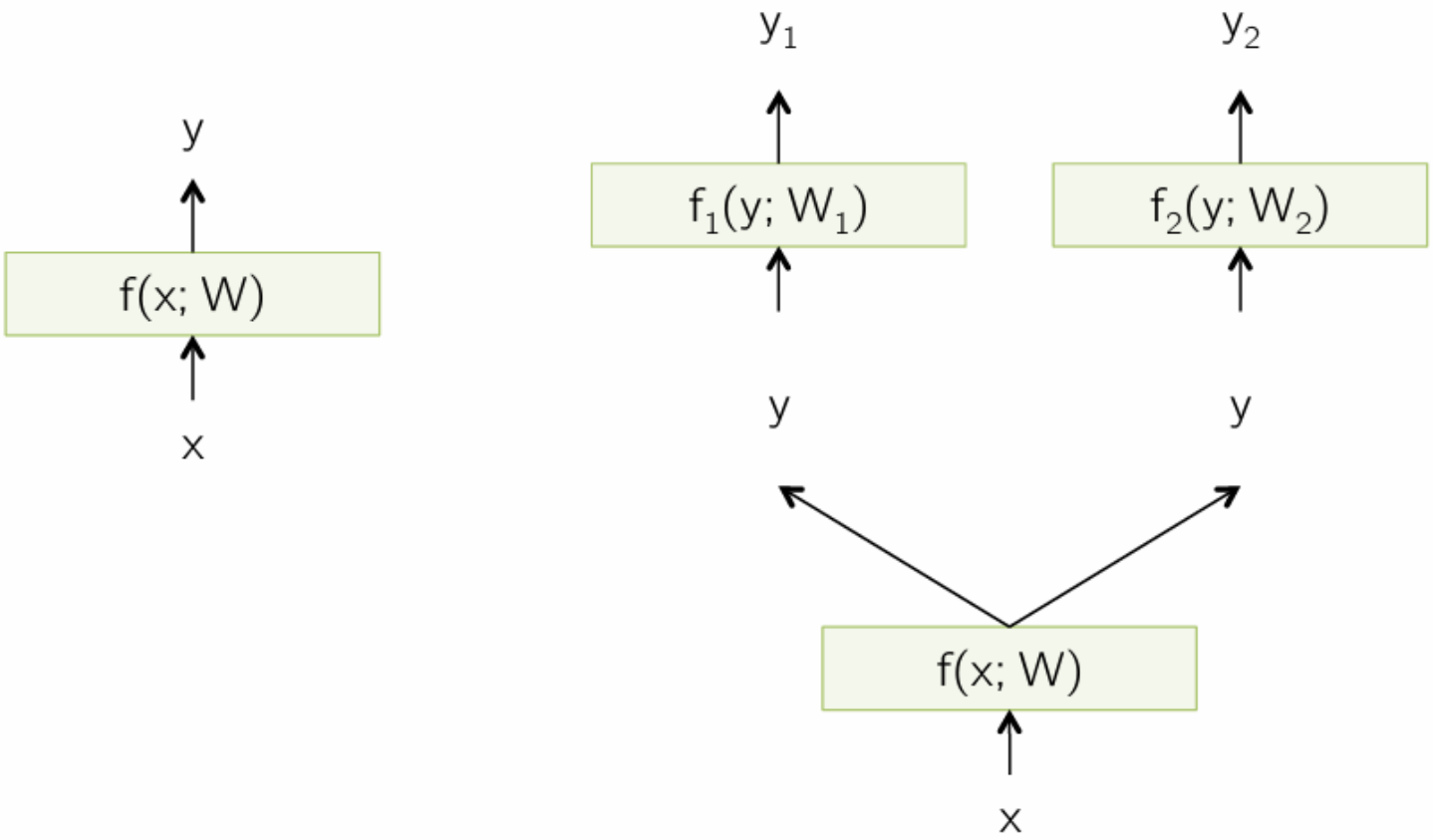
Intrinsic to the layer are:

$\left(\frac{\partial y}{\partial W}\right)$ – How does output change due to params

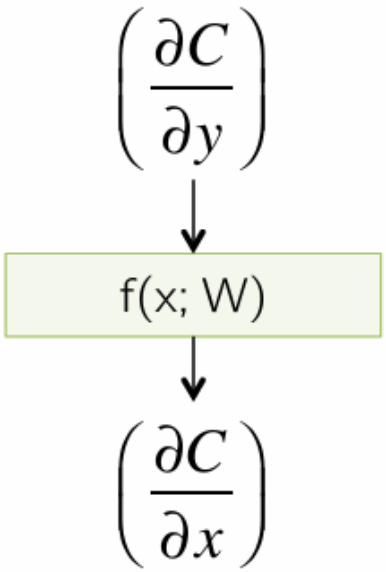
$\left(\frac{\partial y}{\partial x}\right)$ – How does output change due to inputs

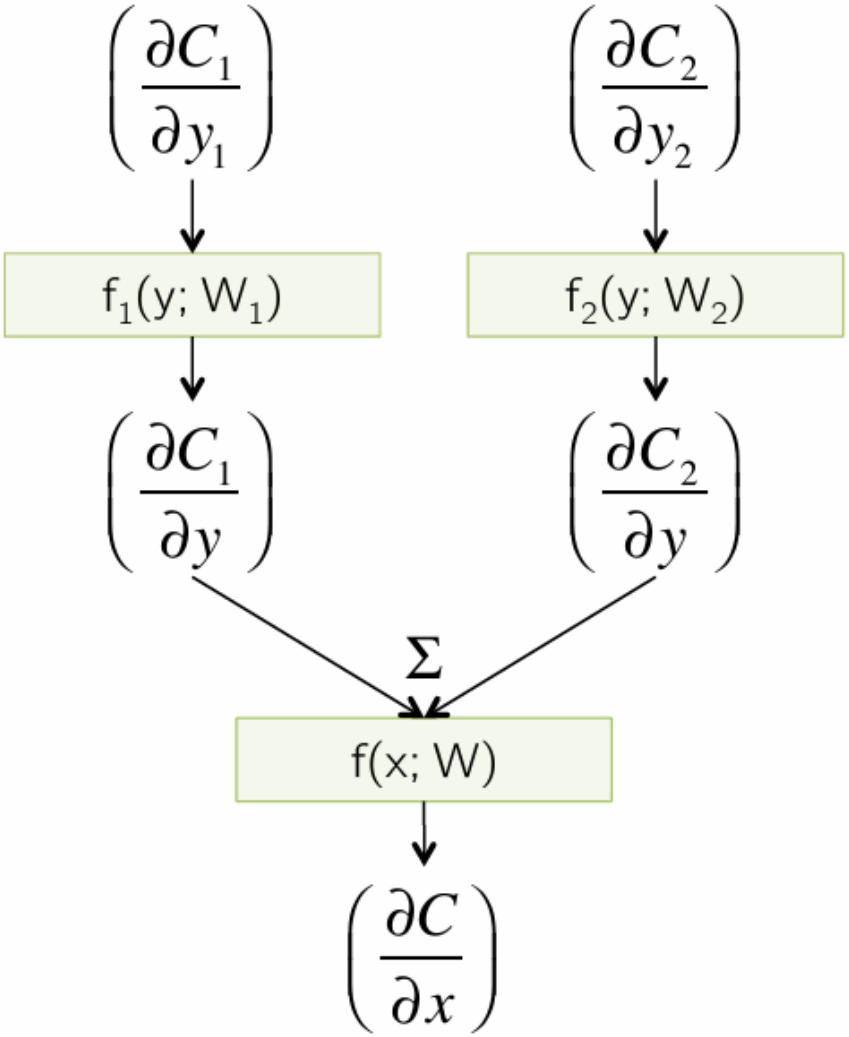
$$\left(\frac{\partial C}{\partial W}\right) = \left(\frac{\partial C}{\partial y}\right)\left(\frac{\partial y}{\partial W}\right) \qquad \left(\frac{\partial C}{\partial x}\right) = \left(\frac{\partial C}{\partial y}\right)\left(\frac{\partial y}{\partial x}\right)$$

Extension to Computational Graphs

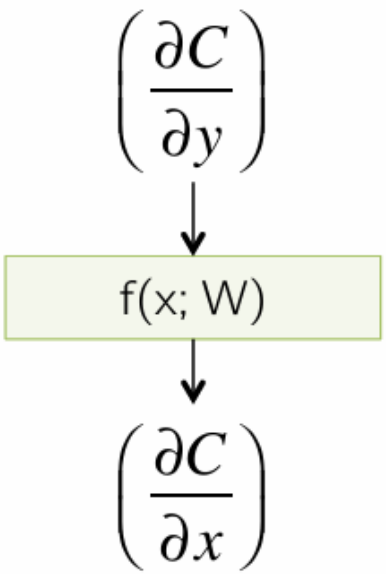


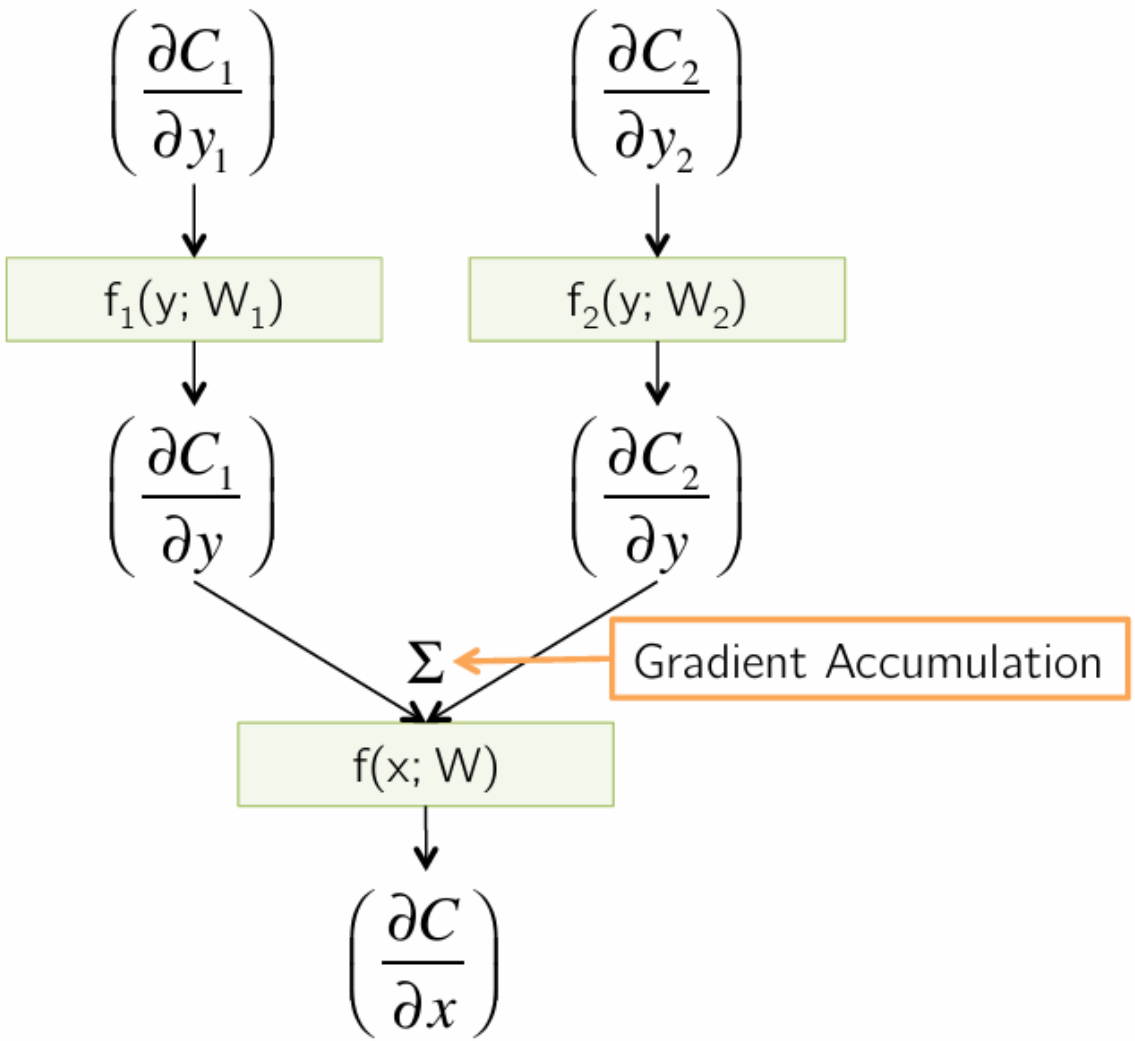
Extension to Computational Graphs





Extension to Computational Graphs





Extension to Computational Graphs

An RNN creates a long computational graph over time:

- Many copies of the same function
- Each depends on previous steps
- All share the same weights W

So gradients must:

1. Flow backward through **every time step**
2. Be **summed** across all contributions
3. Use the **same weights** repeatedly

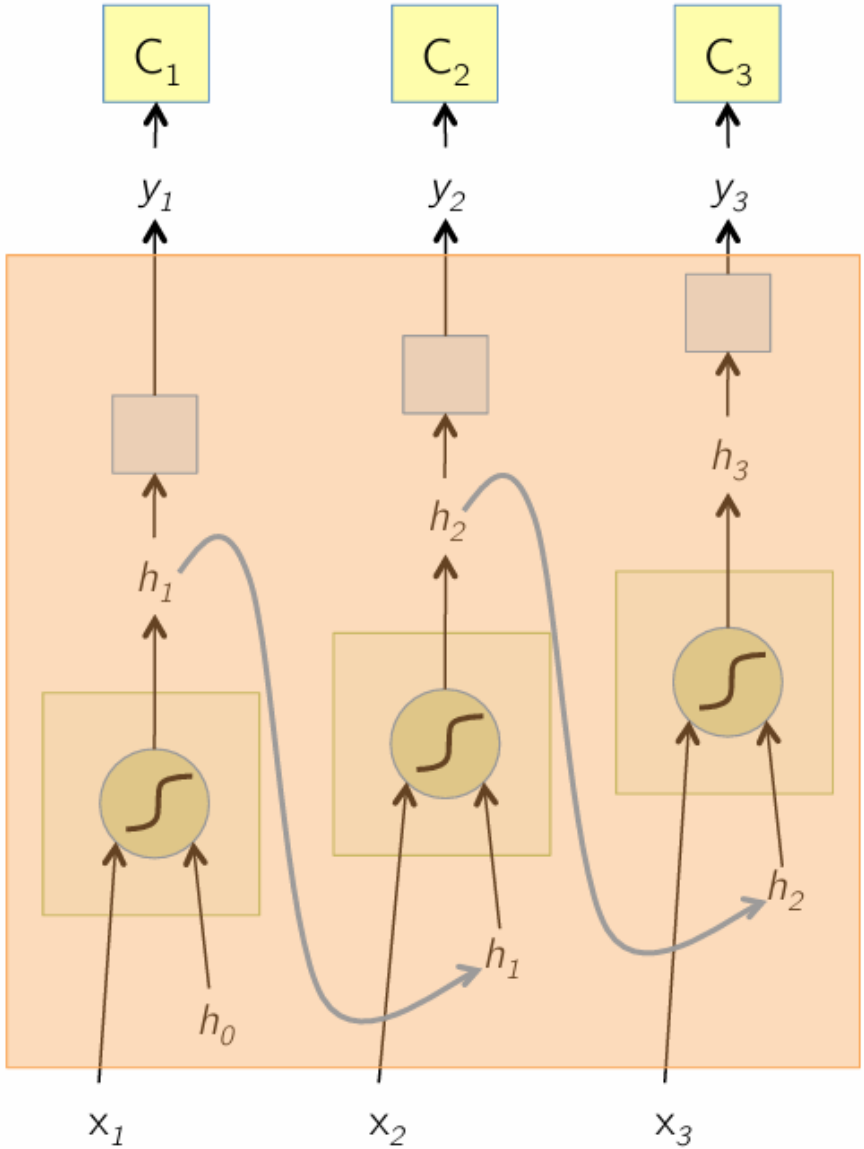
This process is called **Backpropagation Through Time (BPTT)**.

BackPropagation Through Time (BPTT)

- One of the methods used to train RNNs
- The unfolded network (used during forward pass) is treated as one big feed-forward network
- This unfolded network accepts the whole time series as input

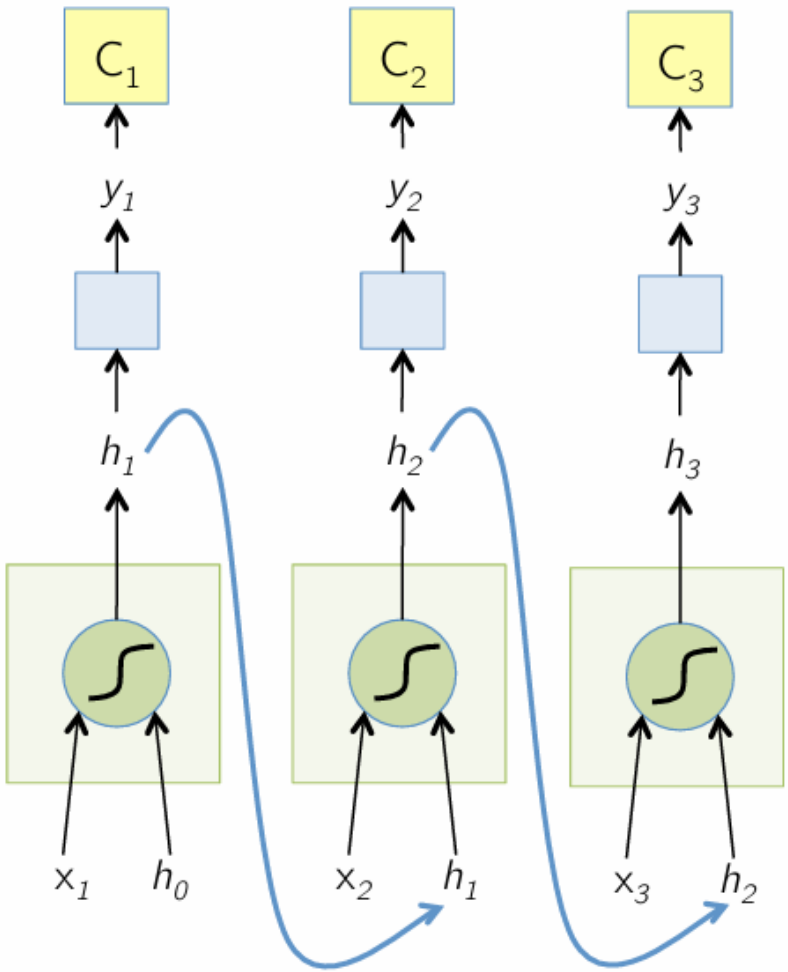
- The weight updates are computed for each copy in the unfolded network, then summed (or averaged) and then applied to the RNN weights

The Unfolded Vanilla RNN

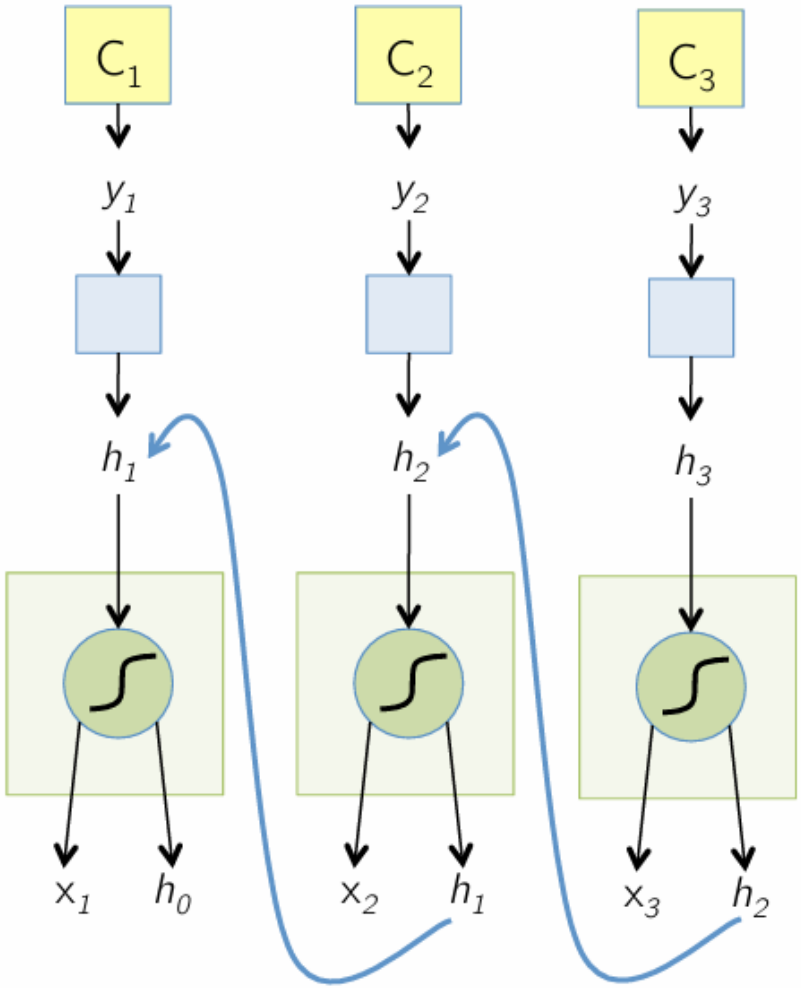


- Treat the unfolded network as one big feed-forward network!
- This big network takes in entire sequence as an input
- Compute gradients through the usual backpropagation
- Update shared weights

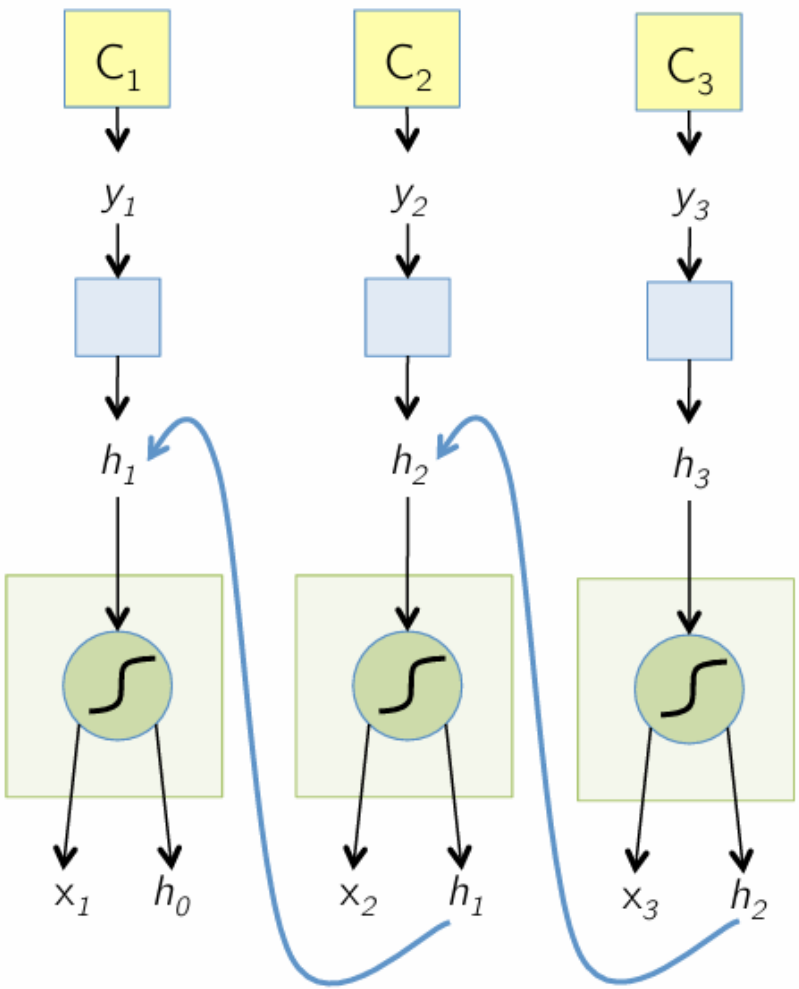
The Unfolded Vanilla RNN Forward



The Unfolded Vanilla RNN Backward



The Vanilla RNN Backward



$$h_t = \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

$$y_t = \mathrm{F}(h_t)$$

$$C_t = \mathrm{Loss}(y_t, \mathrm{GT}_t)$$

$$\frac{\partial C_t}{\partial h_1} = \left(\frac{\partial C_t}{\partial y_t}\right)\left(\frac{\partial y_t}{\partial h_1}\right)$$

$$= \left(\frac{\partial C_t}{\partial y_t}\right)\left(\frac{\partial y_t}{\partial h_t}\right)\left(\frac{\partial h_t}{\partial h_{t-1}}\right)\cdots\left(\frac{\partial h_2}{\partial h_1}\right)$$

Issues with the Vanilla RNNs

- In the same way a product of k real numbers can shrink to zero or explode to infinity, so can a product of matrices

- It is sufficient for $\lambda_1 < 1/\gamma$, where λ_1 is the largest singular value of W, for the **vanishing gradients** problem to occur and it is necessary for **exploding gradients** that $\lambda_1 > 1/\gamma$, where $\gamma = 1$ for the tanh non-linearity and $\gamma = 1/4$ for the sigmoid non-linearity [1]

- Exploding gradients are often controlled with gradient element-wise or norm clipping

The Identity Relationship

- Recall $$\frac{\partial C_t}{\partial h_1} = \left(\frac{\partial C_t}{\partial y_t}\right)\left(\frac{\partial y_t}{\partial h_1}\right)$$
$$= \left(\frac{\partial C_t}{\partial y_t}\right)\left(\frac{\partial y_t}{\partial h_t}\right)\left(\frac{\partial h_t}{\partial h_{t-1}}\right)\cdots\left(\frac{\partial h_2}{\partial h_1}\right)$$

$$h_t = \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$
$$y_t = \mathrm{F}(h_t)$$
$$C_t = \mathrm{Loss}(y_t, \mathrm{GT}_t)$$

- Suppose that instead of a matrix multiplication, we had an **identity relationship** between the hidden states

$$h_t = h_{t-1} + F(x_t)$$
$$\Rightarrow \left(\frac{\partial h_t}{\partial h_{t-1}}\right) = 1$$

- The gradient does not decay as the error is propagated all the way back aka "Constant Error Flow"

What Happens With Normal RNNs?

For a normal RNN:

$$\frac{\partial h_t}{\partial h_{t-1}} = U^\top \cdot diag(1 - tanh^2)$$

- The derivative of tanh is ≤ 1
- U has singular values typically < 1

So the gradient gets multiplied by something <1 **many times**.

→ **Gradient shrinks to zero**

→ **Vanishing gradient problem**

Identity Relationship Fixes This

Now take your simplified update:

$$\boldsymbol{h_t = h_{t-1} + F(x_t)}$$

Differentiate w.r.t. h_{t-1}:

$$\boldsymbol{\frac{\partial h_t}{\partial h_{t-1}} = 1}$$

Because the derivative of h_t with respect to h_{t-1} is literally just 1.

This is the identity.

when computing the gradient from time t back to time 1:

$$\boldsymbol{\frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial h_{t-2}} \cdots \frac{\partial h_2}{\partial h_1}}$$

Each term equals **1**.

Thus:

$$\boldsymbol{\frac{\partial C_t}{\partial h_1} = \frac{\partial C_t}{\partial h_t} \cdot 1 \cdot 1 \cdot 1 \cdots}$$

The gradient **does not shrink** as we go backwards through time.

This Is Called "Constant Error Flow"

Because:

- The error signal travels backward
- Without multiplying by small numbers
- So the gradient remains roughly constant

This is extremely important in RNNs:

- It prevents the vanishing gradient problem.
- It allows long-term memory.
- It is one motivation behind LSTM and GRU design.

LSTMs preserve gradient flow using a similar idea:

$$c_t = c_{t-1} + input$$

The cell state c_t acts like a memory with **identity recurrence**, allowing gradients to flow without vanishing.

The **identity-relationship RNN** (where $h_t = h_{t-1} + F(x_t)$ (solves the *vanishing gradient* problem, but it introduces **several new disadvantages**. These disadvantages are exactly why more advanced RNNs like **LSTM** and **GRU** were invented.
Below are the major drawbacks.

✕ 1. Exploding Hidden States

Since the update is:

$$h_t = h_{t-1} + F(x_t),$$

the hidden state **keeps adding** contributions from all past inputs.
If the sequence is long, even small values accumulate:

$$h_t = F(x_1) + F(x_2) + \cdots + F(x_t)$$

This can cause:

- **Hidden state to blow up** (grow unbounded)
- **Training to become unstable**
- **Exploding gradients** during backpropagation

This is the opposite problem of vanishing gradients.

No Forgetting — Memory Cannot Be Erased

Because of the identity connection:

$$h_t = h_{t-1} + something$$

there is no mechanism to **remove** or **reset** old information.

So:

- Irrelevant past information stays forever
- Hidden state becomes "cluttered"
- Model cannot selectively forget

Real sequence data often needs forgetting (e.g., language, speech).

LSTMs/GRUs add **gates** that allow forgetting.

Hidden State Becomes an Uncontrolled Sum

The hidden state becomes:

- A **running total**
- Not structured
- Hard to interpret
- Hard for the next layers to use

This can reduce model expressiveness.

No Nonlinearity on the State Path

Identity recurrence eliminates the nonlinear transformation on the recurrent path:

$$h_t = h_{t-1} + nonlinear(x_t).$$

This reduces:

- representational power
- ability to learn complex temporal patterns

Normal RNNs (and LSTMs/GRUs) apply nonlinearities to hidden states.

Unstable Long-Term Dynamics

Since h_t is a sum of all previous updates, long-term behavior becomes:

- hard to control
- noisy
- sensitive to small changes in early inputs

The network may learn unstable dynamics over long sequences.

Not Suitable for Most Real Sequence Tasks

This architecture works mostly for:

- synthetic experiments
- very simple sequences
- tasks requiring constant error flow *only*

But for real tasks (language modeling, speech, translation), you need:

- ability to *forget*
- nonlinear transformations
- controlled memory flow

This is why LSTMs and GRUs dominate.

The Identity Relationship

- Recall $$\frac{\partial C_t}{\partial h_1} = \left(\frac{\partial C_t}{\partial y_t}\right)\left(\frac{\partial y_t}{\partial h_1}\right)$$

$$= \left(\frac{\partial C_t}{\partial y_t}\right)\left(\frac{\partial y_t}{\partial h_t}\right)\left(\frac{\partial h_t}{\partial h_{t-1}}\right)\cdots\left(\frac{\partial h_2}{\partial h_1}\right)$$

$$h_t = \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

$$y_t = \mathrm{F}(h_t)$$

$$C_t = \mathrm{Loss}(y_t, \mathrm{GT}_t)$$

- Suppose that instead of a matrix multiplication, we had an **identity relationship** between the hidden states

$$h_t = h_{t-1} + F(x_t)$$

Remember Resnets?

$$\Rightarrow \left(\frac{\partial h_t}{\partial h_{t-1}}\right) = 1$$

- The gradient does not decay as the error is propagated all the way back aka "Constant Error Flow"

Disclaimer

- The explanations in the previous few slides are handwavy

- For rigorous proofs and derivations, please refer to
  On the difficulty of training recurrent neural networks, Pascanu *et al.*, 2013
  Long Short-Term Memory, Hochreiter *et al.*, 1997
  And other sources

Long Short-Term Memory (LSTM)[1]

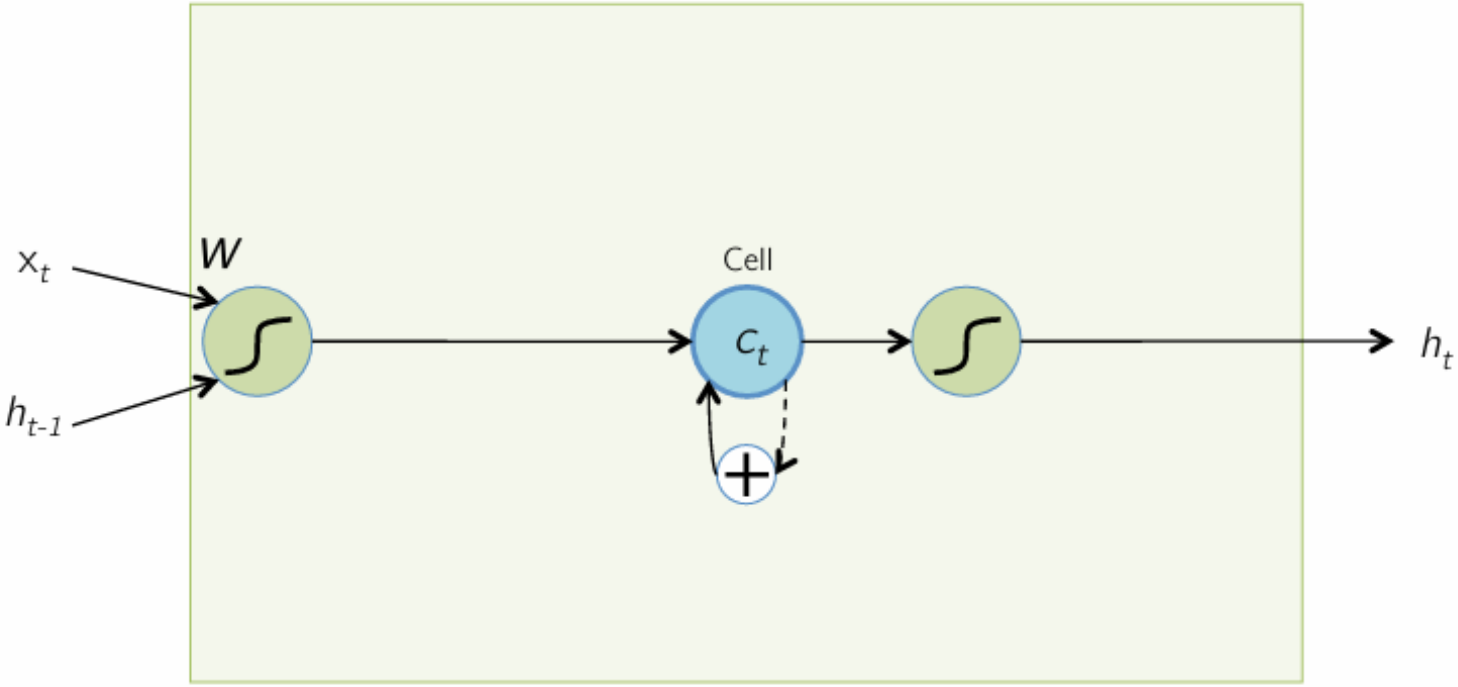
- The LSTM uses this idea of "Constant Error Flow" for RNNs to create a "Constant Error Carousel" (CEC) which ensures that gradients don't decay

- The key component is a memory cell that acts like an accumulator (contains the identity relationship) over time

- Instead of computing new state as a matrix product with the old state, it rather computes the difference between them. Expressivity is the same, but gradients are better behaved

[1] Long Short-Term Memory, Hochreiter *et al.*, 1997

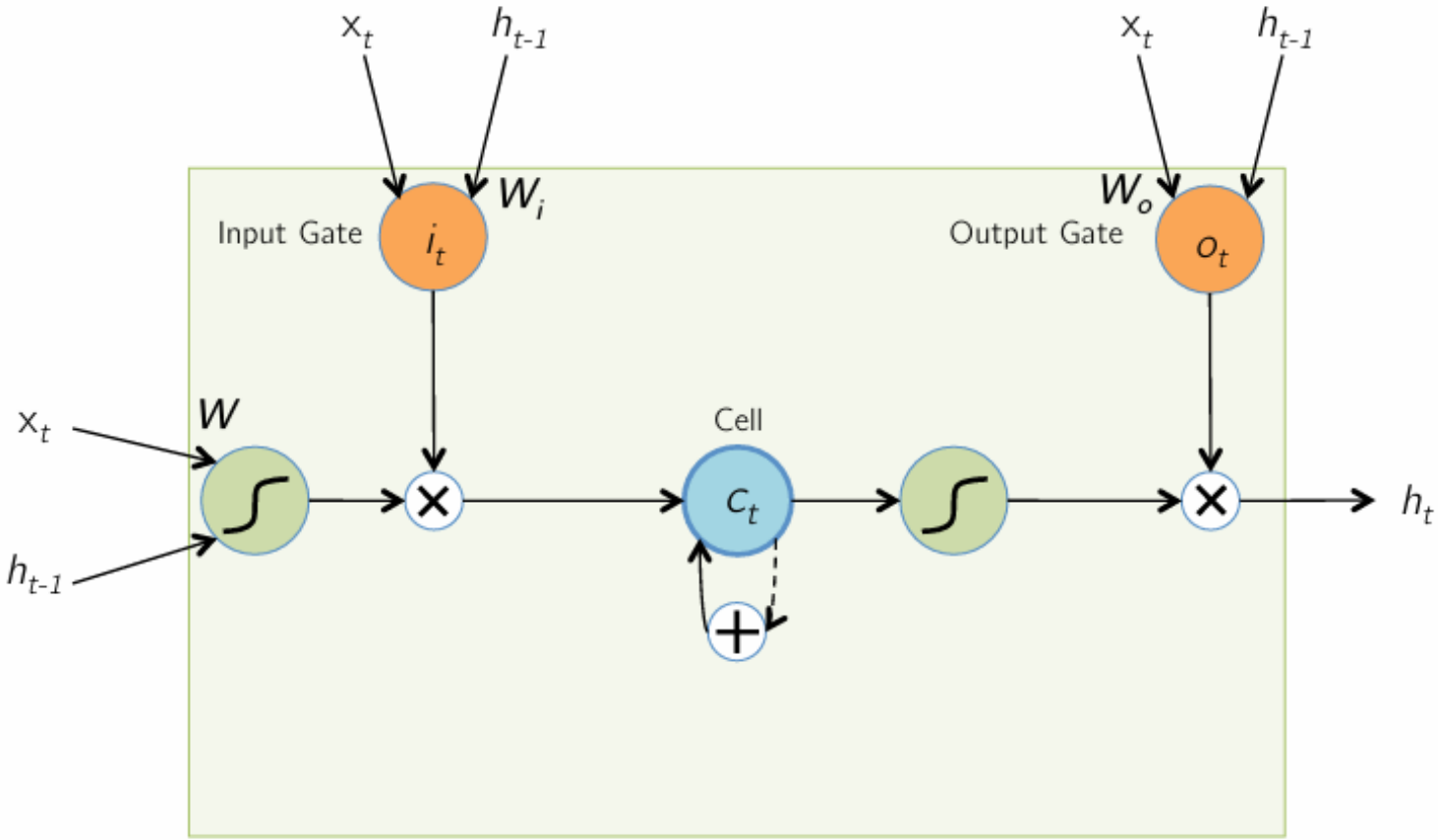
The LSTM Idea



$$c_t = c_{t-1} + \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix} \qquad h_t = \tanh c_t$$

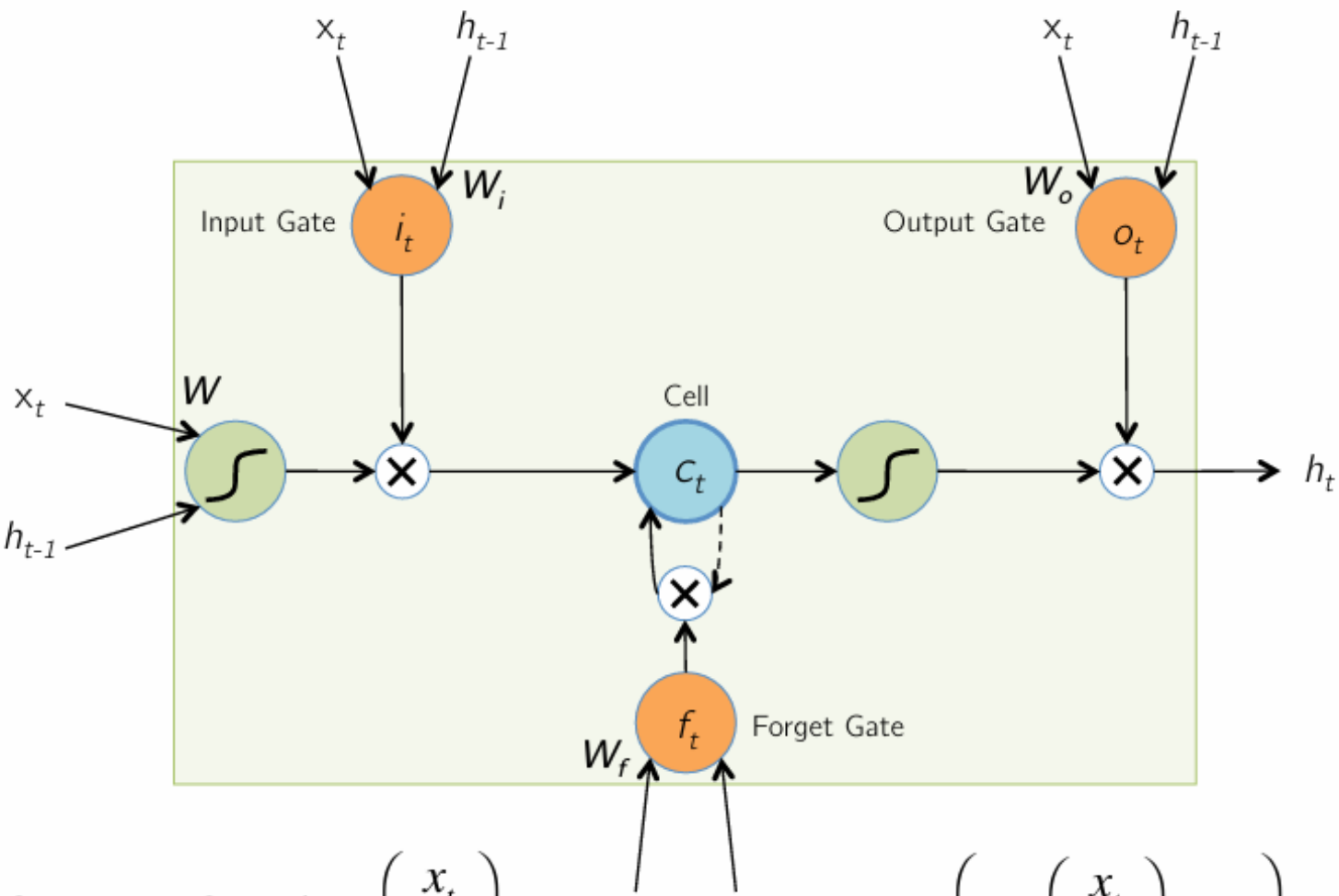
* Dashed line indicates time-lag

The Original LSTM Cell



$$c_t = c_{t-1} + i_t \otimes \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix} \quad h_t = o_t \otimes \tanh c_t \quad i_t = \sigma\left(W_i \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix} + b_i \right)$$ Similarly for o_t

The Popular LSTM Cell



$$c_t = f_t \otimes c_{t-1} + i_t \otimes \tanh W \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix}$$

$$f_t = \sigma\left(W_f \begin{pmatrix} x_t \\ h_{t-1} \end{pmatrix} + b_f \right)$$

Summary

- RNNs allow for processing of variable length inputs and outputs by maintaining state information across time steps
- Various Input-Output scenarios are possible (Single/Multiple)

- Vanilla RNNs are improved upon by LSTMs which address the vanishing gradient problem through the CEC
- Exploding gradients are handled by gradient clipping

- More complex architectures are listed in the course materials for you to read, understand, and present

Other Useful Resources / References

- http://cs231n.stanford.edu/slides/winter1516_lecture10.pdf
- http://www.cs.toronto.edu/~rgrosse/csc321/lec10.pdf

- R. Pascanu, T. Mikolov, and Y. Bengio, On the difficulty of training recurrent neural networks, ICML 2013
- S. Hochreiter, and J. Schmidhuber, Long short-term memory, Neural computation, 1997 9(8), pp.1735-1780
- F.A. Gers, and J. Schmidhuber, Recurrent nets that time and count, IJCNN 2000
- K. Greff , R.K. Srivastava, J. Koutník, B.R. Steunebrink, and J. Schmidhuber, LSTM: A search space odyssey, IEEE transactions on neural networks and learning systems, 2016
- K. Cho, B. Van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, Learning phrase representations using RNN encoder-decoder for statistical machine translation, ACL 2014
- R. Jozefowicz, W. Zaremba, and I. Sutskever, An empirical exploration of recurrent network architectures, JMLR 2015